



DCA0305

morsinaldo.medeiros.075@ufrn.edu.br
ivanovitch.silva@ufrn.br

Inside the Transformer

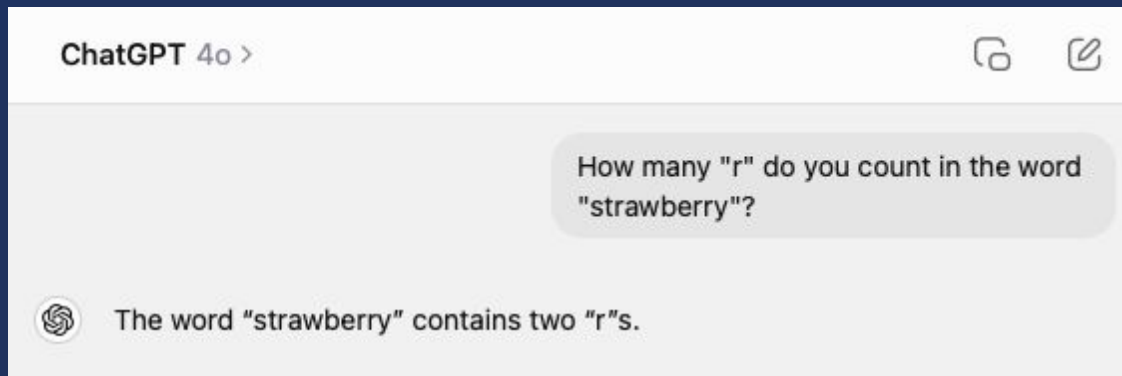
From tokens to sampling

Agenda



How many “r”s in strawberry?

For years, frontier models answered **2**.

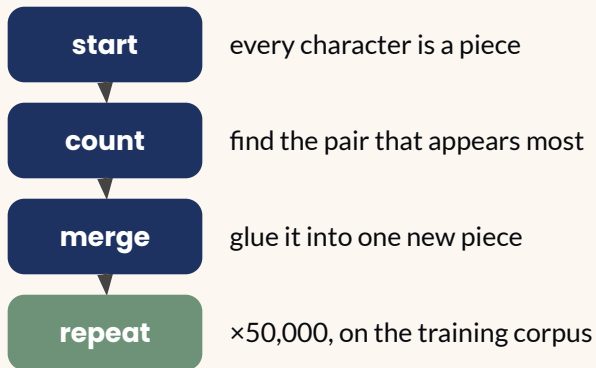


Tokens: text becomes pieces

50,257

pieces in GPT-2's vocabulary

How the pieces are chosen (BPE)



Tiktokenizer

gpt2

strawberry

Token count
3

strawberry

[301, 1831, 8396]

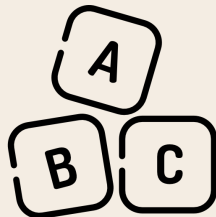
Why not words? Why not letters?

Words



- Vocabulary explodes (every inflection)
- “farduddle”? unknown token
- Special characters (<eos>)

Letters



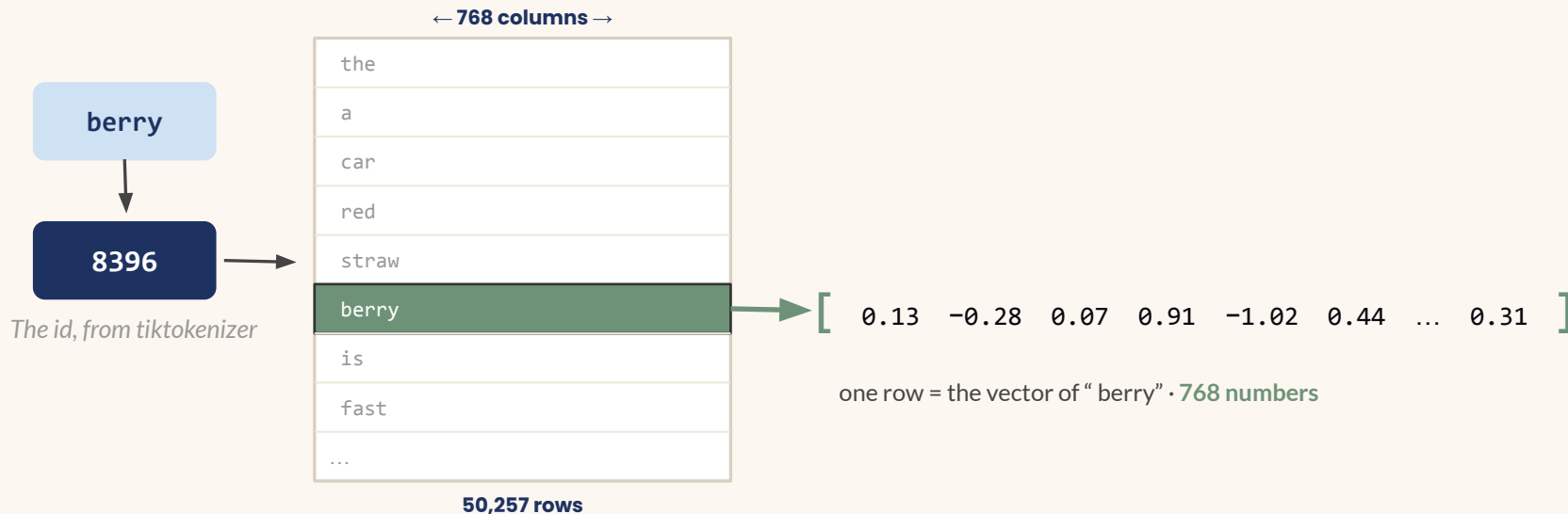
- Nothing is ever unknown
- But sequences get longer
- And attention pays length²

Subwords

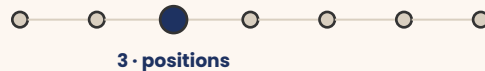


- Common words = one piece
- Rare words = a few pieces
- Fixed vocab, nothing unknown

Embeddings: the id becomes a vector



This table alone: $50,257 \times 768 \approx 38.6\text{M}$ of GPT-2's 124M parameters



Positions: attention has no order

“a red car” = “car red a”

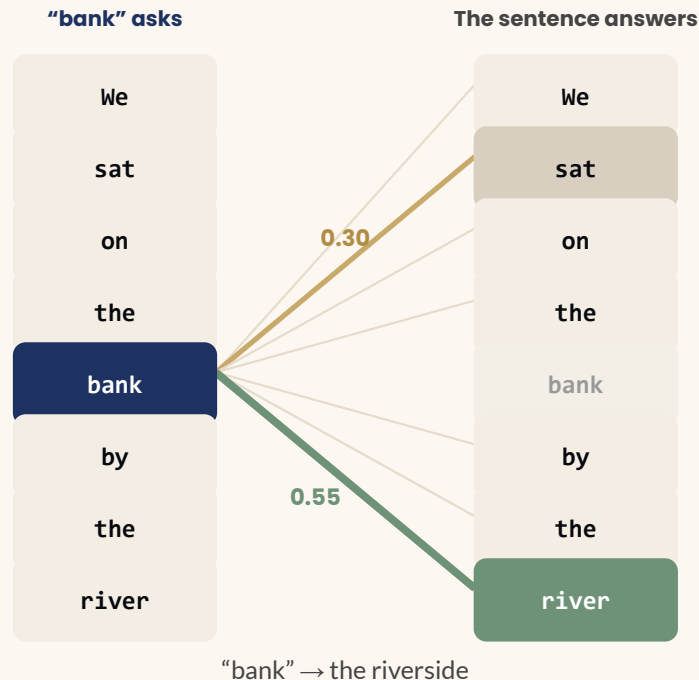
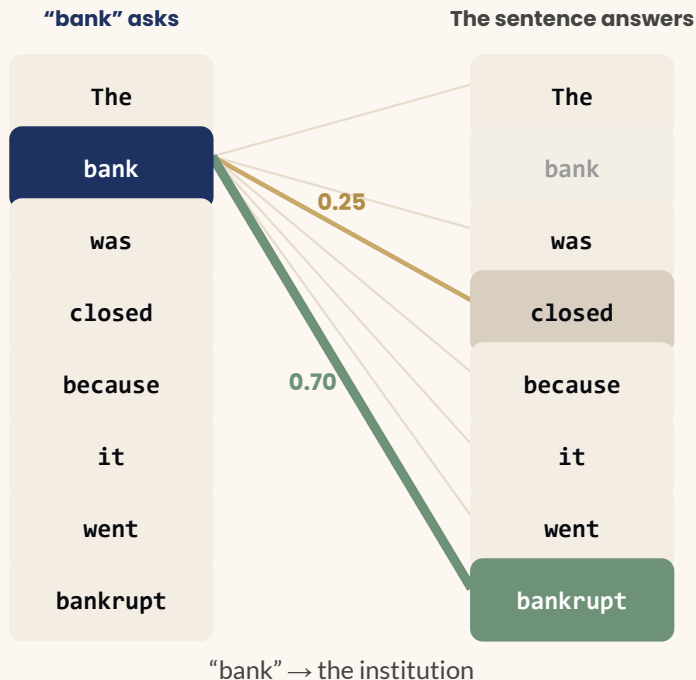
...for attention alone: it scores tokens as an unordered set. in language, order is meaning.

The fix: give each position its own vector, and add it, number by number

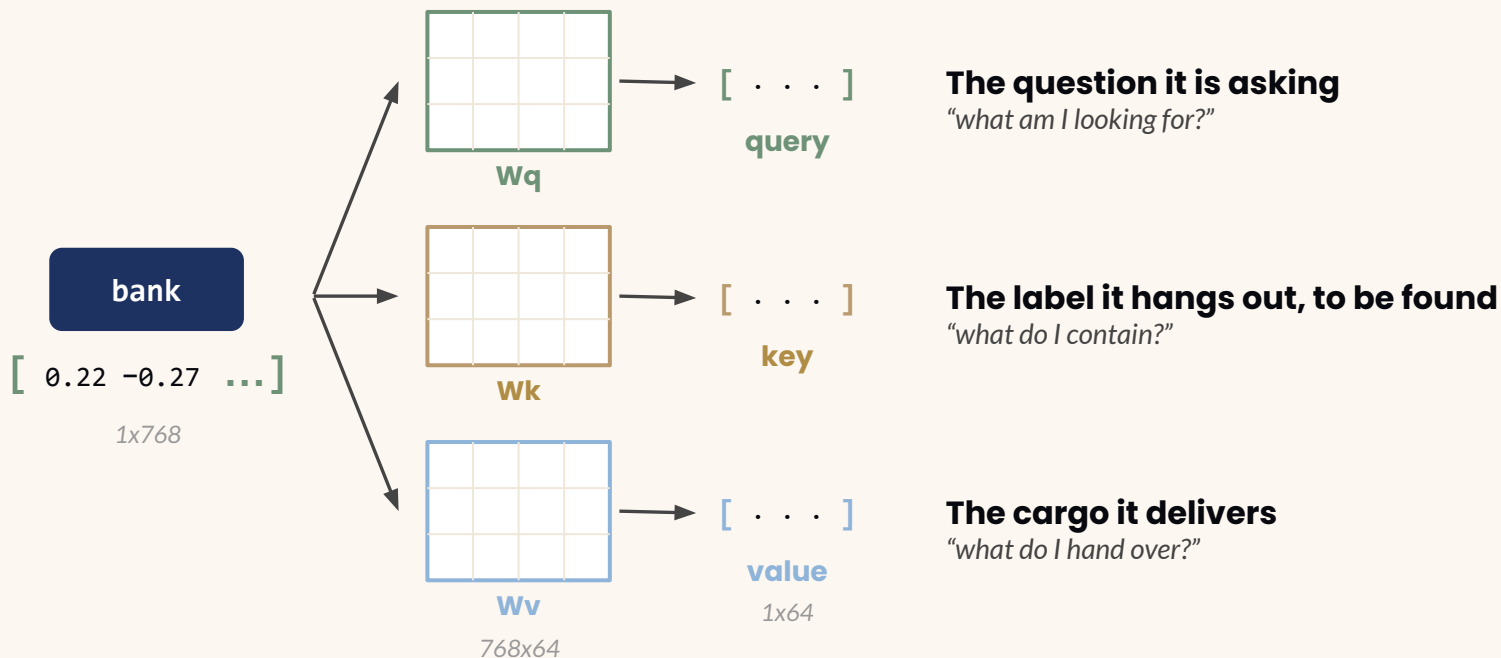
a	$\begin{bmatrix} 0.21 & -0.05 & 0.33 & \dots \end{bmatrix}$	+	pos 0	$\begin{bmatrix} 0.02 & 0.11 & -0.07 & \dots \end{bmatrix}$	=	$\begin{bmatrix} 0.23 & 0.06 & 0.26 & \dots \end{bmatrix}$	a @ 0
red	$\begin{bmatrix} -0.14 & 0.48 & 0.09 & \dots \end{bmatrix}$	+	pos 1	$\begin{bmatrix} 0.05 & -0.03 & 0.12 & \dots \end{bmatrix}$	=	$\begin{bmatrix} -0.09 & 0.45 & 0.21 & \dots \end{bmatrix}$	red @ 1
car	$\begin{bmatrix} 0.13 & -0.28 & 0.07 & \dots \end{bmatrix}$	+	pos 2	$\begin{bmatrix} 0.09 & 0.01 & -0.20 & \dots \end{bmatrix}$	=	$\begin{bmatrix} 0.22 & -0.27 & -0.13 & \dots \end{bmatrix}$	car @ 2

Token vector (what) + **Position vector** (where) = **What enters the attention layers**

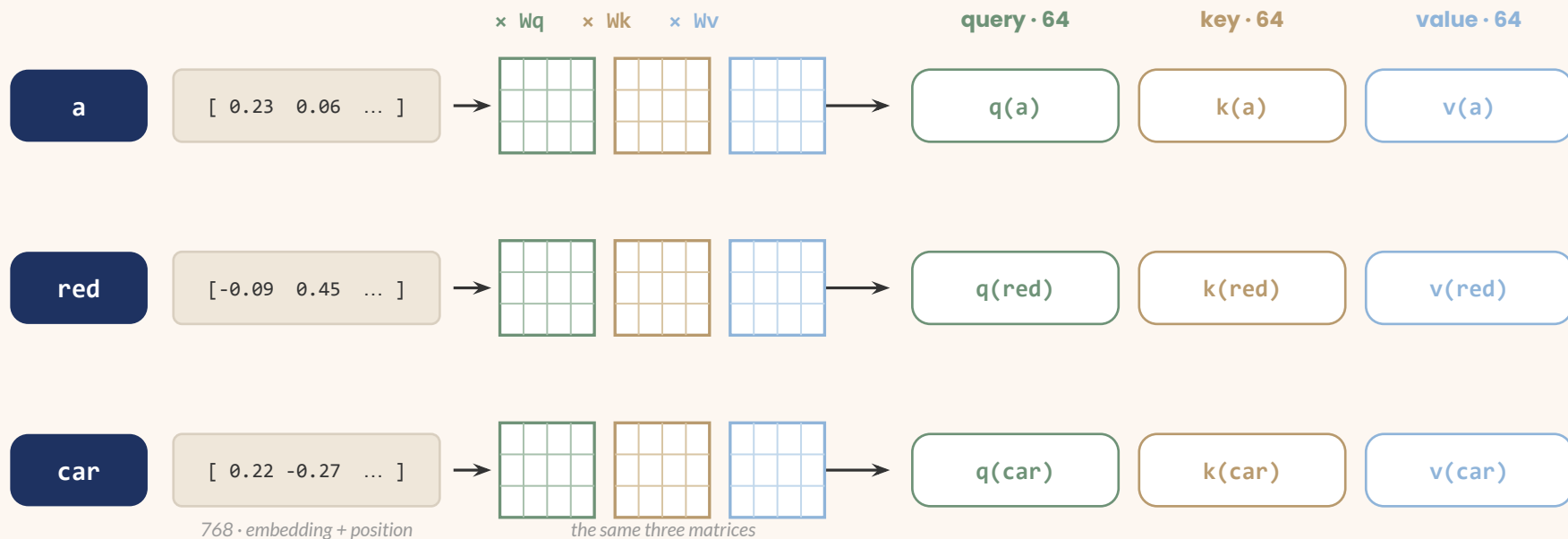
Attention: a word decides what it means by looking around



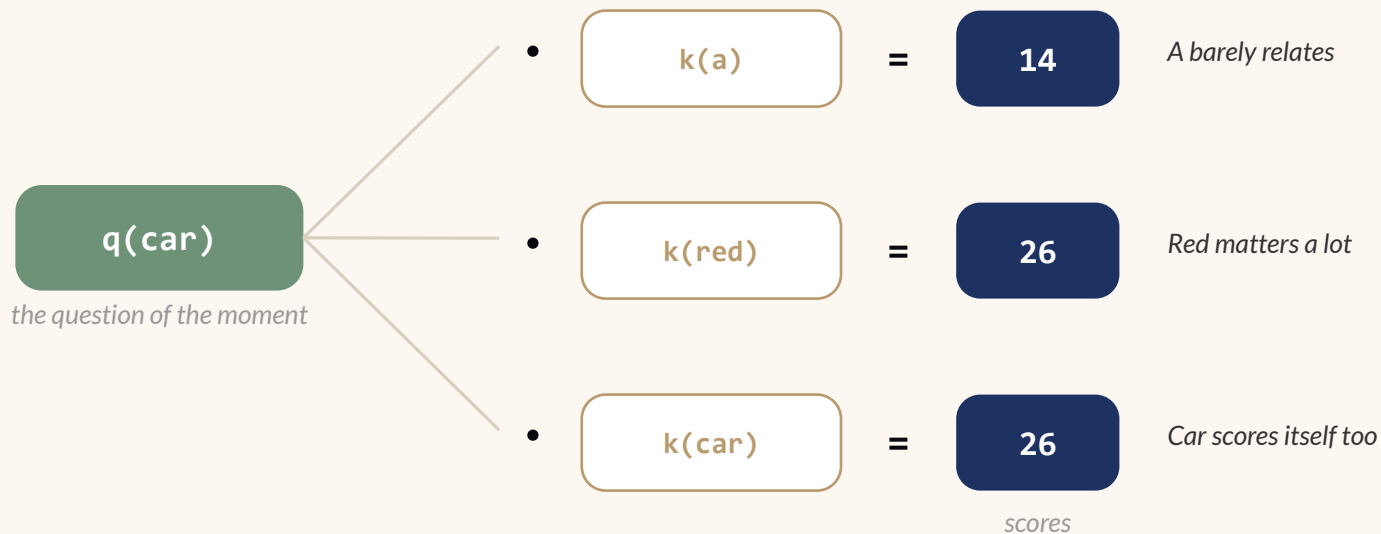
Every token asks, offers, and delivers: Q, K, V



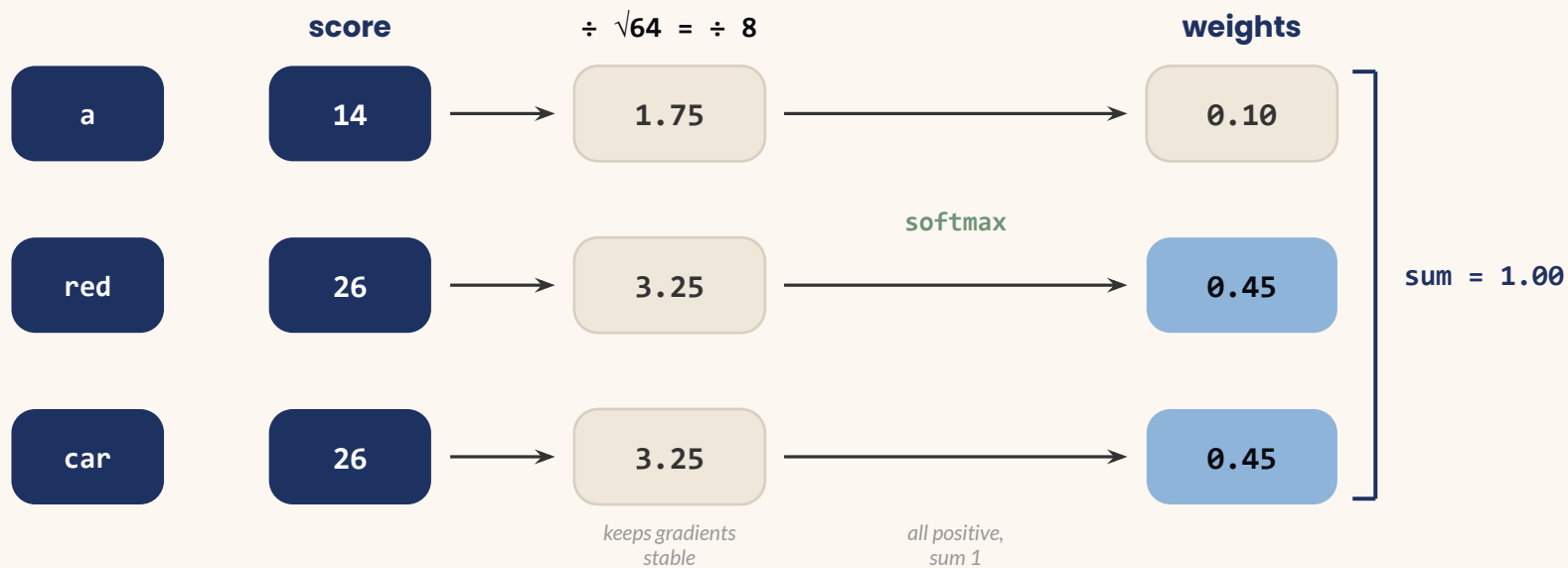
Step 1 · every token gets a q, a k, and a v



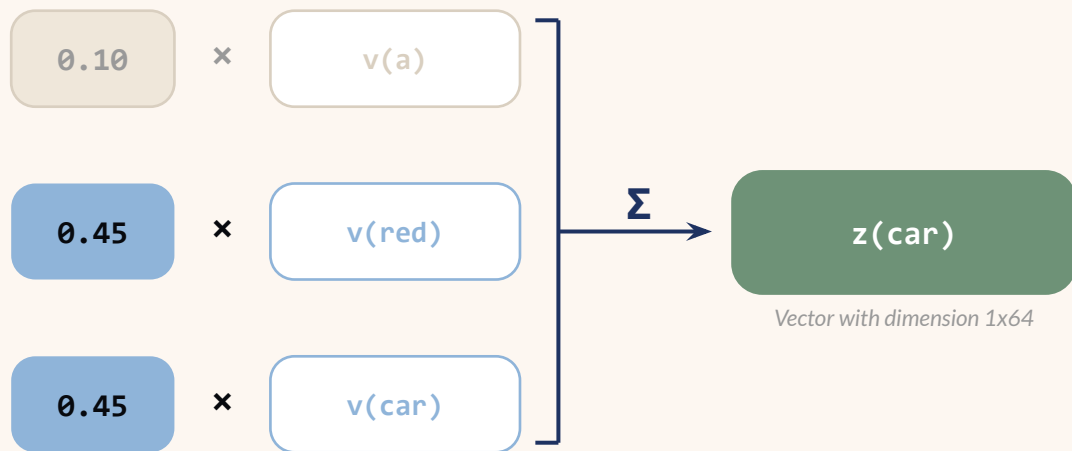
Step 2 · car asks, every key answers



Steps 3 and 4 · scale, then softmax



Steps 5 and 6 · weight every value, then sum



$z(\text{car})$ is what leaves the attention layer for this token



One head, in matrices

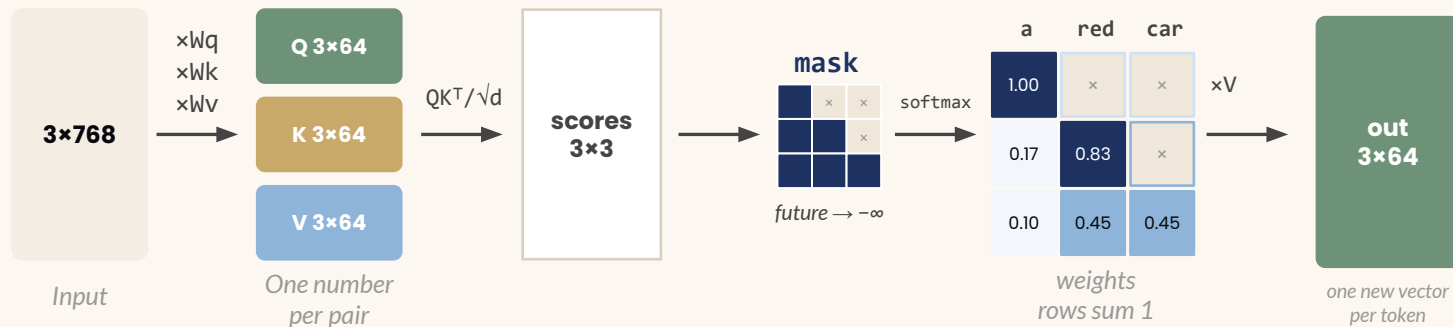
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

QK^T all questions $\times$ all labels, at once · step 2

$\sqrt{d_k}$ keeps the softmax tame · step 3

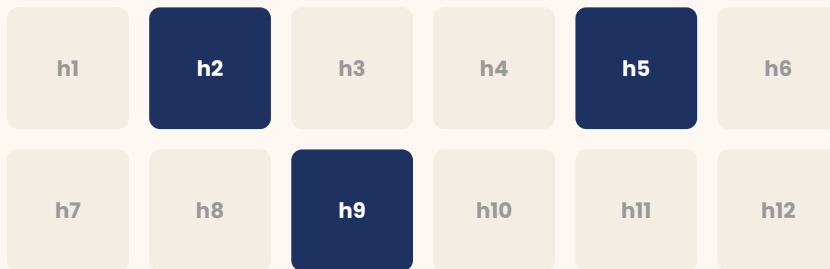
softmax rows become weights, sum 1 · step 4

V the mix · steps 5 and 6



Multi-head: twelve ways of looking, at once

$$12 \text{ heads} \times 64 \text{ dims} = 768$$



h2 tracks adjacency, “red” next to “car”

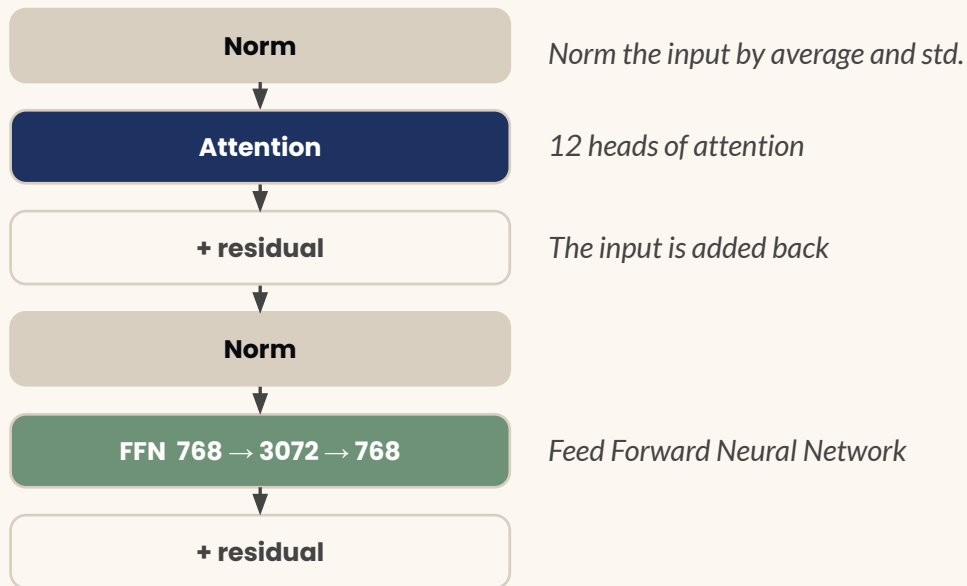
h5 resolves reference, “ele” → “banco”

h9 follows syntax, verb → subject

concat the 12 outputs → one projection → 768 again

The block: attention, then a private think

×12 identical blocks in GPT-2

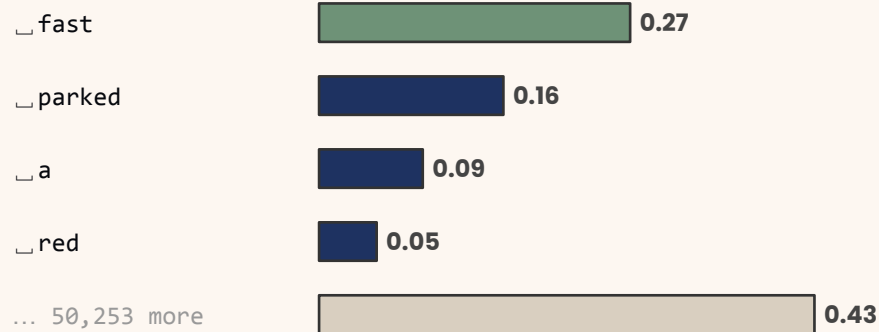


The output is a distribution, not an answer

50,257 numbers · Sum = 1.0000

After the last block, one score per vocabulary piece. Softmax makes them a distribution.

Next token after “a red car is...” (illustrative)



$T \approx 0$

Always the top bar

$T = 0.7$

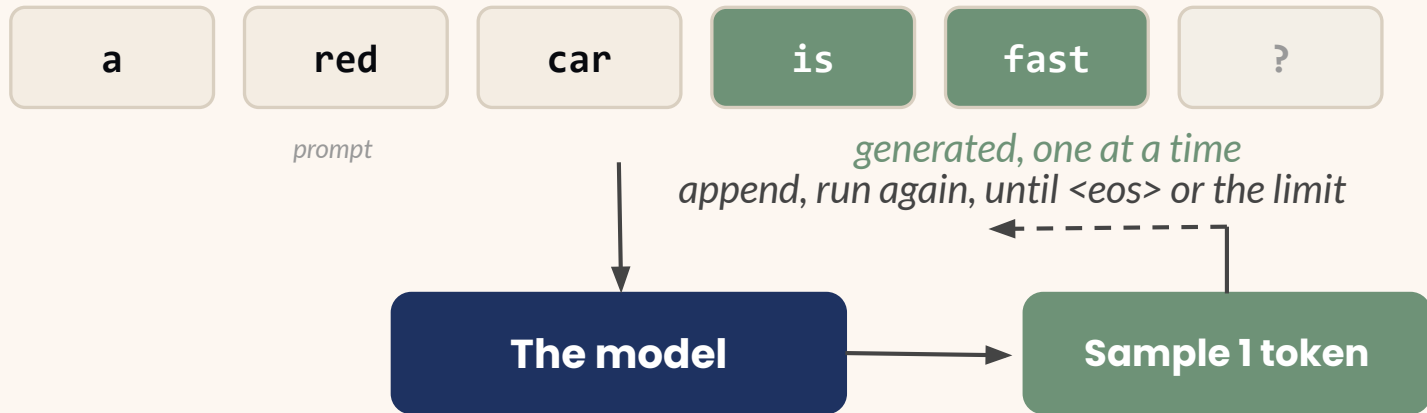
Mostly the big bars

$T = 1.5$

Long tail wakes up

This is why the same question gets different answers.

The loop: one token, append, repeat





7 · the model

The whole model, counted by hand

124M parameters · GPT-2 small



embeddings $50,257 \times 768 = 38.6M$

12 blocks $\times \sim 7.1M = 85M$

Positions: the sliver of 0.8M

The sum: $38.6 + 0.8 + 85 \approx 124M$

Clone me

<https://github.com/ivanovitchm/aiengineering>

- **Inside the Transformer: from tokens to sampling**

- Follow **"a red car"** through GPT-2 (124M), from tokenization to generation.
- **Seven stops:** tokens, embeddings, positions, attention, Transformer blocks, sampling, and the full model.
- Builds self-attention step by step, including multi-head attention and causal masking.
- Closes by breaking down GPT-2's **124M parameters**.

- **Notebook** **Open in Colab** **The seven stops, in code**

- Replays the lecture using the real **GPT-2 small** from Hugging Face.
- Explores tokenization, embeddings, positions, attention, Transformer blocks, sampling, and generation.
- Implements **one attention head by hand** and verifies it against the library output.
- Runs entirely on the **free Colab CPU**, with no training required.

- **Notebook** **Open in Colab** **Exercise: a model selection report**

- Compare **GPT-2, SmolLM2-360M, and Qwen2.5-0.5B** for an application in your own domain.
- Evaluate **tokenization/context, architecture, attention, classification, generation, and inference cost**.
- Use ≥ 30 labelled examples for zero- and few-shot classification.
- **Deliverable:** executed notebook with evidence-based decisions, a final comparison table, and a **3-minute presentation**.